

Notes: Atanasov, Julio, and Leng (RFS, 2024)

The Bright Side of Political Uncertainty: The Case of R&D

Contents

1	Big picture	3
2	Data and measurement (key objects)	4
2.1	Gubernatorial elections and treatment definition	4
2.2	Firm-level data and sample construction	4
2.3	Key variables	5
2.4	Variables used to test mechanisms	5
3	Models and empirical specifications (all equations + interpretation)	6
3.1	Theoretical motivation: bad-news principle versus growth options	6
3.2	Baseline annual differences-in-differences specification	6
3.3	Quarterly specification and comparison with capital expenditures	6
3.4	Cross-sectional heterogeneity specifications	7
3.5	Dynamic specification over the election cycle	7
3.6	Alternative uncertainty measure and IV-style design	7
4	Identification and instruments (what makes the estimates credible)	8
4.1	What the design is using	8
4.2	Why close elections proxy political uncertainty	8
4.3	What would threaten identification	8
4.4	Why the mechanism evidence matters	9
4.5	Robustness and instruments	9
5	Tables and figures	9
5.1	Table 1: Summary statistics and validation objects	9
5.2	Table 2: Baseline annual R&D regressions	10
5.3	Table 3: Quarterly R&D versus capital expenditures	10

5.4	Table 4: Close versus nonclose elections and polling validation	10
5.5	Table 5: Politically sensitive industries	11
5.6	Table 6: High-tech firms and growth-option logic	11
5.7	Table 7: Dynamics over the four-year election cycle	12
5.8	Table 8: Asset redeployability and substitution tests	12
5.9	Table 9: State-level EPU index as an alternative proxy	13
5.10	Table 10: Robustness and political-business-cycle concerns	14
6	Short summary	14
7	Conclusion	14
7.1	Core conclusion (what the paper establishes)	14
7.2	Economic intuition	14
7.3	Takeaway	15

1 Big picture

- **Question.** Does political uncertainty always reduce firm investment, or can it *increase* investment when the investment takes the form of **R&D**? More specifically, what is the causal effect of political uncertainty on firm-level R&D spending?
- **Setting.** The paper studies U.S. firms around **close gubernatorial elections**, using them as quasi-natural experiments for political uncertainty. The main annual sample covers **1976–2021** Compustat data and the quarterly sample covers **1988–2021**. The election data span **598 gubernatorial elections** between 1976 and 2020, of which **138** are classified as close.
- **Core challenge.** The standard real-options view says uncertainty should reduce investment because firms value the option to wait when investment is irreversible. But that logic may not apply to R&D. R&D can instead resemble a **growth option**: an early-stage investment that creates the possibility of a much larger future payoff if later conditions are favorable.
- **Main contribution.** The paper brings this theoretical distinction to the data. It shows that **political uncertainty raises R&D** even though earlier work had shown that the same type of uncertainty lowers **capital expenditures**. The paper therefore argues that the effect of uncertainty depends on the **type of investment**.
- **Why it matters.** A large literature treats political uncertainty as uniformly harmful for real activity. This paper argues that such a conclusion is incomplete. If some investments are option-like and multi-stage, uncertainty may accelerate them rather than delay them.
- **Main empirical question.** When firms experience heightened political uncertainty because a gubernatorial election is unusually close, do they spend more or less on R&D?
- **Main empirical results.**
 - In the baseline annual specification, close-election years increase **R&D intensity** by about **0.0023–0.0025**.
 - Relative to the sample mean, this implies roughly a **2.63%** increase in R&D intensity; relative to the median, about **6.46%**.
 - In quarterly data, R&D rises significantly in **quarters 2, 3, and 4** of a close-election year, while **capital expenditures fall**, especially in quarters 2 and 3.
 - The effect is much stronger in **close elections** than in nonclose elections, and stronger when the **incumbent is not reelected**.
 - The effect is stronger for **politically sensitive industries** and for **high-tech / high-growth-option firms**.
 - The data do **not** support a simple substitution story in which firms merely cut physical investment and redirect freed cash into R&D.
 - Using the **state-level Economic Policy Uncertainty (EPU)** index gives similar conclusions: R&D rises and capital expenditures fall when policy uncertainty is higher.
- **Economic takeaway.** Political uncertainty can have a **bright side**. It discourages irreversible physical investment but can encourage R&D when that spending buys firms an option to expand later, especially in settings with high technological uncertainty and long development lags.

2 Data and measurement (key objects)

2.1 Gubernatorial elections and treatment definition

- **Primary election data source.** The main source is the **CQ Press Voting and Elections Collection**, supplemented with hand-collected information from Wikipedia.
- **Why gubernatorial elections?**
 - Their timing is largely **fixed by law**, unlike many elections internationally.
 - They occur in **different states and different years**, creating useful cross-state and time-series variation.
 - State governors influence regulation, taxation, labor policy, customer protection, environmental policy, and industry policy, all of which can matter for firms' operating conditions.
- **Election timing details.**
 - Most states hold elections every **4 years**.
 - Vermont and New Hampshire hold them every **2 years**.
 - Five states elect governors in odd-numbered years just before presidential elections.
- **Close election definition.** An election is **close** if the margin of victory, defined as the vote-share difference between the winner and runner-up, is **less than or equal to 5%**.
- **Election statistics.**
 - Total gubernatorial elections in the sample: **598**.
 - Close elections: **138**.
 - Mean victory margin overall: **16.13%**.
 - Mean margin among close elections: **2.59%**.
 - Elections where the incumbent is not reelected: **317**.
- **Close-election cycle.** The empirical design centers on a 4-year cycle

$$[-1, 0, +1, +2],$$

where year 0 is the election year. The treated observations are firm-years in the **close election year**, while the control observations are the years before and after within the same close-election cycle.

2.2 Firm-level data and sample construction

- **Annual firm data.** Compustat North America Fundamentals Annual, **1976–2021**.
- **Quarterly firm data.** Compustat quarterly files, **1988–2021**.
- **Listed and OTC firms.** The sample includes domestic firms listed on **NYSE, AMEX, and NASDAQ**, and also **OTC-traded firms**, which are often smaller technology firms.
- **Size filter.** Firms are required to have total assets of at least **\$5 million**.

- **Sample size.**
 - Annual close-election sample: **26,262 firm-year observations.**
 - Quarterly close-election sample: **112,333 firm-quarter observations.**
- **State assignment.** Firms are matched to elections using the **headquarter state** reported in Compustat.
- **Potential measurement issue.** Compustat reports current headquarters rather than historical headquarters. The paper argues relocations are rare and therefore mostly add noise rather than bias. It also shows robustness to an alternative state-exposure measure based on **García and Norli (2012)** using 10-K textual mentions of states.

2.3 Key variables

- **Main dependent variable. R&D intensity,** defined as

$$\frac{\text{R\&D expenditure (XRD)}}{\text{Total assets (AT)}}.$$

- **Capital-investment comparison variable.** Capital expenditures scaled by assets:

$$\frac{\text{CAPX}}{\text{AT}}.$$

- **Firm controls.** Tobin's q , cash flow, $\ln(\text{Age})$, $\ln(\text{Sales})$, tangibility, profitability, leverage.
- **Industry controls.** The Herfindahl index over three-digit SIC industries, plus its square to allow for inverted-U competition effects.
- **State controls.** State GDP growth and state unemployment rate.
- **Validation variable.** Annualized stock return volatility from CRSP daily returns.

2.4 Variables used to test mechanisms

- **Politically sensitive industries (PSI).** Indicator for firms in tobacco, pharmaceuticals, health care services, defense, petroleum and natural gas, telecommunications, and transportation.
- **High-tech / growth-option proxies.**
 - **BFP High-tech** following Brown, Fazzari, and Petersen (2009): drugs, office equipment and computers, electronic components, communication equipment, scientific and medical instruments, and software.
 - **HJT High-tech** following Hall, Jaffe, and Trajtenberg (2005): pharmaceuticals, medical instruments, chemicals, computers, communications, and electrical industries.
- **Asset redeployability.** Industry-level redeployability measure from Kim and Kung (2017), used to test whether the R&D result is mainly driven by substitution away from physical capital.
- **Alternative uncertainty measure. State-level EPU index** from Baker, Davis, and Levy (2022), based on about **3,500 local newspapers.**

3 Models and empirical specifications (all equations + interpretation)

3.1 Theoretical motivation: bad-news principle versus growth options

- **Standard irreversible-investment view.** In Bernanke (1983)-style models, uncertainty raises the option value of waiting, so firms delay irreversible investment. This is the paper's benchmark for why **capital expenditures** should fall around uncertain elections.
- **Growth-option view of R&D.** In models such as Bar-Ilan and Strange (1996, 1998) and Pindyck (1993), R&D is often a **multi-stage investment**. An early-stage outlay creates an option to continue later or abandon if the project goes badly. Because the downside is limited but the upside expands when uncertainty is high, the paper's prediction is that political uncertainty can **raise** R&D.
- **Economic interpretation.** The same uncertainty shock can lower one type of investment and raise another, depending on whether the investment is mainly irreversible physical capital or an exploratory growth option.

3.2 Baseline annual differences-in-differences specification

The paper's main annual specification is

$$\text{R\&D Intensity}_{ijt} = \alpha_i + \gamma_t + \beta_0 \text{CloseElectionDummy}_{jt} + X_{it}\phi + \varepsilon_{ijt},$$

where:

- i indexes firms,
- j indexes states,
- t indexes years,
- α_i are firm fixed effects,
- γ_t are year fixed effects,
- $\text{CloseElectionDummy}_{jt} = 1$ in election year t if state j has a close gubernatorial election,
- X_{it} contains firm controls and, in some specifications, state macro controls.

Interpretation. The coefficient β_0 measures how the same firm's R&D intensity changes in a close-election year relative to non-election years within the close-election cycle. The design is therefore a within-firm event-study style DD exploiting staggered close elections across states.

3.3 Quarterly specification and comparison with capital expenditures

Using quarterly data, the paper estimates

$$\text{R\&D Intensity}_{ijt} = \alpha_i + \gamma_t + \sum_{k=1}^4 \beta_k \text{CloseElectionQuarter}_{kt} + X_{it}\phi + \varepsilon_{ijt},$$

where $\text{CloseElectionQuarter}_{kt}$ indicates quarters 1 through 4 of a close-election year.

The same quarterly design is also estimated with

$$\frac{\text{CAPX}}{\text{AT}}$$

as the dependent variable.

Interpretation. This comparison isolates whether political uncertainty affects all investment similarly. The paper finds the opposite: R&D rises in quarters 2–4, while capital expenditures fall, especially in quarters 2 and 3.

3.4 Cross-sectional heterogeneity specifications

The paper then augments the baseline design with subgroup splits and interaction terms:

- **Close versus nonclose election cycles.**
- **Incumbent reelected versus not reelected.**
- **Politically sensitive versus non-sensitive industries.**
- **High-tech versus low-tech firms.**
- **Asset redeployability interactions.**

A generic interacted version is of the form

$$\text{R\&D Intensity}_{ijt} = \alpha_i + \gamma_t + \beta_0 \text{CloseElectionDummy}_{jt} + \beta_1 (\text{CloseElectionDummy}_{jt} \times Z_{it}) + X_{it}\phi + \varepsilon_{ijt},$$

where Z_{it} is a characteristic such as being politically sensitive, high-tech, or having more redeployable assets.

Interpretation. These regressions test mechanisms. If the growth-option channel is right, the election effect should be stronger where R&D is more option-like: politically sensitive sectors, high-tech sectors, and sectors with higher technological uncertainty.

3.5 Dynamic specification over the election cycle

To study pre- and post-election dynamics, the paper estimates

$$\text{R\&D Intensity}_{ijt} = \alpha_i + \gamma_t + \beta_0 \text{Preelection}(-1)_{jt} + \beta_1 \text{CloseElection}(0)_{jt} + \beta_2 \text{PostElection}(+1)_{jt} + \beta_3 \text{PostElection}(+2)_{jt} + \varepsilon_{ijt},$$

Interpretation. This checks whether the close-election R&D result is driven by pretrends or by a post-election rebound. Unlike capital expenditures in earlier work, R&D does not display a clear post-election boom. The R&D increase is concentrated in the **close election year itself**.

3.6 Alternative uncertainty measure and IV-style design

The paper also estimates quarterly regressions using

$$\log(\text{State EPU Index})$$

as the uncertainty proxy. In one specification, the election-quarter indicators are used as instruments for the state EPU index, isolating the component of newspaper-based uncertainty that is associated with close elections.

Interpretation. This serves as both an external-validity check and a way to verify that the close-election design is indeed picking up policy uncertainty rather than some other state-level shock.

4 Identification and instruments (what makes the estimates credible)

4.1 What the design is using

- **Exogenous timing.** The timing of gubernatorial elections is set by law and not determined by contemporaneous firm-level conditions.
- **Close elections as uncertainty shocks.** The paper treats highly contested elections as shocks to uncertainty about future policy rather than simply policy changes themselves.
- **Staggered variation.** Elections occur in different states and different years, giving a staggered panel structure that helps separate election uncertainty from nationwide shocks.
- **Within-firm identification.** Firm fixed effects absorb time-invariant firm characteristics; year or quarter fixed effects absorb aggregate shocks.

4.2 Why close elections proxy political uncertainty

- **Validation with stock volatility.** Stock return volatility is higher in election quarters than in nonelection quarters, and higher in close elections than in nonclose elections.
- **Validation with polling.** Ex post election closeness is strongly related to ex ante polling consensus: when the race is more of a “tossup,” the ex post margin is smaller and the probability of a close election is higher.
- **Alternative closeness proxy.** Results are stronger when the **incumbent is not reelected**, another setting in which uncertainty is plausibly greater.

4.3 What would threaten identification

- **State business cycles.** The concern is that close elections may coincide with state-specific economic conditions that directly affect R&D. The paper addresses this with state-level GDP growth and unemployment controls.
- **Political business cycles in spending.** If governors increase spending in uncertain election years, firms might react to spending rather than uncertainty. The paper tests this directly and finds no significant increase in total state expenditures or expenditures relative to revenues in election years.
- **Headquarter-state mismeasurement.** Compustat headquarters may be noisy. The results remain robust using a time-varying state exposure measure based on 10-K textual mentions.

4.4 Why the mechanism evidence matters

- **Growth-option evidence.** The election effect is concentrated in politically sensitive and high-tech firms, exactly where R&D is more likely to be multi-stage and to embed growth options.
- **Substitution evidence against a simple cash reallocation story.** If firms merely cut capex and reallocate cash into R&D, then the effect should be stronger among firms with low asset redeployability. The paper does *not* find this for R&D, although it does find the redeployability pattern for capital expenditures. That makes substitution an incomplete explanation.
- **No post-election catch-up in R&D.** Capital expenditures often show a bust-and-boom pattern around uncertainty. R&D does not. This is more consistent with firms taking on new exploratory projects than with simply postponing or reallocating standard investment.

4.5 Robustness and instruments

- Triple clustering by state, year, and firm.
- Exchange-listed firms only, excluding OTC firms.
- Excluding the dot-com period and the global financial crisis.
- Excluding California, Massachusetts, and New York.
- Alternative R&D scaling by sales.
- Alternative state assignment from 10-Ks.
- International replication using the Julio and Yook election framework.
- Restricting to innovative firms with at least one patent.
- Using close-election quarters as instruments for state EPU.

5 Tables and figures

5.1 Table 1: Summary statistics and validation objects

Read:

- The paper has **598** gubernatorial elections, of which **138** are close.
- The annual firm sample has **26,262** firm-year observations; the quarterly sample has **112,333** firm-quarter observations.
- Election quarters are associated with higher stock return volatility than nonelection quarters, and close elections are associated with higher volatility than nonclose elections.

Takeaway. Table 1 both describes the sample and validates the basic idea that elections, especially close ones, are periods of elevated uncertainty.

Table 1

Summary statistics

A. Election characteristics

	N	Mean	Median	SD
Gubernatorial elections	598			
Incumbent Republican	287			
Incumbent Democrat	303			
Incumbent other	8			
Victory margin (%)	598	16.13	12.67	13.52
Close election (%)	138	2.59	2.65	1.43
Incumbent reelected	281			

B. Annual sample firm characteristics (close election sample)

	N	Mean	Median	SD
R&D intensity	26,262	0.0876	0.0356	0.1503
q	26,262	2.2201	1.5143	2.4729
Cash flow	26,262	-0.0389	0.0652	0.3727

C. Quarterly sample firm characteristics

	N	Mean	Median	SD
R&D intensity	112,333	0.0121	0.0000	0.0265
Capital expenditures/assets	112,333	0.0157	0.0088	0.0209
q	112,333	2.2623	1.5279	3.3986
Cash flow	112,333	0.0017	0.0188	0.0957

D. Average annualized realized stock return volatility

Nonelection quarter	0.4192			
Election quarter	0.4614			
Nonclose elections		0.4552		
Close elections		0.4687		
t-test of difference	33.995	4.972		

Panel A reports summary statistics for gubernatorial elections held between 1975 and 2020. Panel B reports summary statistics for the annual firm-level data in the analysis. Panel C reports summary statistics the quarterly firm-level data. Panel D reports the annual mean R&D intensity during nonelection quarters, election quarters, and close-election quarters. See the [appendix](#) for variable descriptions as well as the variable sources.

Table 2

Political uncertainty and R&D intensity: Baseline results

		Dependent variable: R&D intensity					
		(1)	(2)	(3)	(4)	(5)	(6)
Close election dummy	0.0025*** (0.0009)	0.0025*** (0.0009)	0.0025*** (0.0009)	0.0023*** (0.0009)	0.0023*** (0.0009)	0.0023*** (0.0011)	0.0023*** (0.0011)
q	0.0040*** (0.0008)	0.0040*** (0.0008)	0.0040*** (0.0008)	0.0040*** (0.0008)	0.0040*** (0.0008)	0.0040*** (0.0011)	0.0040*** (0.0011)
Cash flow	-0.0416*** (0.0148)	-0.0416*** (0.0148)	-0.0416*** (0.0148)	-0.0416*** (0.0148)	-0.0416*** (0.0148)	-0.0416*** (0.0163)	-0.0416*** (0.0163)
ln(Sales)	0.0014 (0.0016)	0.0014 (0.0016)	0.0014 (0.0016)	0.0014 (0.0016)	0.0014 (0.0016)	0.0014 (0.0016)	0.0014 (0.0016)
ln(Age)	0.0101** (0.0043)	0.0101** (0.0043)	0.0101** (0.0043)	0.0101** (0.0043)	0.0101** (0.0043)	0.0101** (0.0045)	0.0101** (0.0045)
Tangibility	0.1054*** (0.0147)	0.1053*** (0.0147)	0.1053*** (0.0147)	0.1053*** (0.0147)	0.1053*** (0.0147)	0.1053*** (0.0149)	0.1053*** (0.0149)
Profitability	-0.2244*** (0.0114)	-0.2245*** (0.0114)	-0.2245*** (0.0114)	-0.2245*** (0.0114)	-0.2245*** (0.0114)	-0.2245*** (0.0151)	-0.2245*** (0.0151)
Leverage	-0.0262** (0.0127)	-0.0262** (0.0127)	-0.0262** (0.0127)	-0.0262** (0.0127)	-0.0262** (0.0127)	-0.0262** (0.0129)	-0.0262** (0.0129)
Herfindahl		-0.0039 (0.0025)	-0.0039 (0.0025)	-0.0038 (0.0026)	-0.0038 (0.0026)	-0.0038 (0.0039)	-0.0038 (0.0039)
GDP growth				0.0425 (0.0276)	0.0425 (0.0276)	0.0425 (0.0340)	0.0425 (0.0340)
Unemployment				-0.0001 (0.0008)	-0.0001 (0.0008)	-0.0001 (0.0008)	-0.0001 (0.0008)
N	25,177	25,177	25,177	25,177	25,177	25,177	25,177
R ²	.8447	.8447	.8447	.8447	.8447	.8447	.8447
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

The dependent variable in all regressions is R&D intensity, defined as the ratio of R&D expenditure to total assets. The main explanatory variable is the close election year dummy (year 0). The sample is over close election periods (margin of victory is less than 5%). Table A1 in the [appendix](#) defines the variable descriptions as well as the variable sources in detail. Standard errors, reported in parentheses below coefficient estimates, are clustered at the state level following [Bertrand, Duflo, and Mullainathan \(2004\)](#) in columns 1–4, at the state and firm levels in column 5, and at the state and year levels in column 6, and corrected for heteroscedasticity. Data are for the period 1976 to 2020. * $p < .1$; ** $p < .05$; *** $p < .01$.

5.2 Table 2: Baseline annual R&D regressions

Read:

- The close-election dummy is positive and significant in every specification.
- The baseline coefficient is about **0.0023**.
- This corresponds to an increase of about **2.63%** relative to the mean R&D intensity and about **6.46%** relative to the median.

Takeaway. The core result is not fragile to the inclusion of firm controls, state controls, or alternative clustering choices.

5.3 Table 3: Quarterly R&D versus capital expenditures

Read:

- R&D rises significantly in **quarters 2, 3, and 4** of a close-election year.
- Capital expenditures fall significantly in **quarters 2 and 3**, with the strongest decline in quarter 3.

Takeaway. This is one of the sharpest pieces of evidence in the paper: the same uncertainty shock pushes R&D and capital expenditures in **opposite directions**.

5.4 Table 4: Close versus nonclose elections and polling validation

Read:

- The election effect on R&D is large and significant for **close elections** but economically small and insignificant for **nonclose elections**.

Table 3
Differences between R&D and fixed investment: Quarterly data

	(1)	(2)	(3)	(4)
Dependent variable	R&D intensity	R&D intensity	Capex/Assets	Capex/Assets
Close election quarter 1	0.0001 (0.0003)	0.0001 (0.0003)	−0.0001 (0.0002)	−0.0001 (0.0002)
Close election quarter 2	0.0014*** (0.0003)	0.0013*** (0.0003)	−0.0006*** (0.0002)	−0.0005** (0.0002)
Close election quarter 3	0.0012*** (0.0003)	0.0011*** (0.0003)	−0.0008*** (0.0002)	−0.0007*** (0.0002)
Close election quarter 4	0.0012*** (0.0003)	0.0012*** (0.0003)	−0.0003 (0.0002)	−0.0002 (0.0002)
N	112,333	112,333	112,333	112,333
R ²	.2075	.2498	.1848	.1938
Firm controls	No	Yes	No	Yes
Firm FE	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes

This table reports investment regressions for both R&D and capital expenditures using Compustat quarterly data. The sample is over close election periods (margin of victory is less than 5%). Close Election election quarter dummies for quarters 1 through 4 are set equal to one during the specified quarter in an election year if the election is classified as "close." Table A1 in the appendix defines the variable descriptions as well as the variable sources in detail. The variables of interest are the close election quarter indicators. We control for firm and quarter fixed effects and in some specifications below coefficient estimates, are clustered at the firm and quarter levels. Data are for the period 1988 to 2021. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 4
R&D in close versus nonclose election periods

A. R&D regressions				
	Dependent variable: R&D intensity			
	Close elect (1)	Nonclose elect (2)	Gov not reelected (3)	Gov reelected (4)
Election dummy	0.0023*** (0.0009)	0.0007 (0.0005)	0.0017** (0.0008)	−0.0002 (0.0010)
N	25,177	70,200	48,872	46,014
R ²	.8447	.8267	.8362	.8331
Firm controls	Yes	Yes	Yes	Yes
State controls	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
B. Ex ante and ex post measures of election closeness				
	Margin of victory (1)	Margin of victory (2)	Close election indicator (3)	Close election indicator (4)
Mean polling consensus	5.1015*** (0.5071)		−0.1517*** (0.0220)	
Median polling consensus		4.3627*** (0.4657)		−0.1371*** (0.0204)
N	673	673	673	673
R ²	.7557	.7431	.5219	.5300

Panel A of this table examines whether the degree of electoral uncertainty amplifies the effect of political uncertainty on firm-level R&D intensity. In columns 1 and 2, we use election closeness to proxy for the degree of electoral uncertainty. Specifically, an election is classified as close if the margin of victory, defined as the vote difference between the first and second place candidates, is less than 5%. In columns 3 and 4, we use an indicator for whether an incumbent governor is reelected to proxy for electoral uncertainty. The dependent variable in all regressions is R&D intensity, defined as the ratio of R&D expenditure to total assets. The regressions include firm and year fixed effects. Standard errors are clustered at the firm level and corrected for heteroscedasticity. Standard errors are reported in square parentheses. To save space, we suppress the estimates of firm-specific and state economics control variables. * $p < .1$; ** $p < .05$; *** $p < .01$.

- The effect is also significant when the incumbent is **not reelected** and essentially zero when the incumbent is reelected.
- Ex ante polling consensus strongly predicts ex post election closeness.

Takeaway. The result is really about **uncertainty intensity**, not just about elections per se.

5.5 Table 5: Politically sensitive industries

Read:

- Politically sensitive industries show an election-year R&D effect of about **0.0090**.
- This implies roughly a **4.12%** increase relative to their mean R&D intensity.
- The increase is about **nine times larger** than in politically nonsensitive industries.

Takeaway. The effect is strongest exactly where political change is most relevant for firm profitability.

5.6 Table 6: High-tech firms and growth-option logic

Read:

- Under the BFP classification, high-tech firms show a close-election coefficient of about **0.0047**; low-tech firms show essentially zero.
- Under the HJT classification, high-tech firms show a coefficient of about **0.0066**; low-tech firms again show essentially zero.
- The interacted coefficients are positive and significant in both definitions.

Table 5
Politically sensitive industries

	Dependent variable: R&D intensity		
	Politically sensitive (1)	Nonsensitive (2)	Interacted (3)
Close election dummy	.0090** (.0042)	.0010 (.0007)	.0007 (.0008)
Close election × PSIs			.0099** (.0039)
N	3,654	21,523	25,177
R ²	.8376	.8443	.8448
Firm controls	Yes	Yes	Yes
State controls	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

This table examines the cross-sectional variations of politically sensitive industries on firm-level R&D intensity in election years. Specifically, *politically sensitive industries (PSIs)* is an indicator variable set equal to one if firms fall into the following industries: Tobacco Products (5), Pharmaceuticals (13), Health Care Services (11), Defense (26), Petroleum and Natural Gas (30), Telecommunications (32) and Transportation (40), as used in [Herron et al. \(1999\)](#) and [Julio and Yook \(2012\)](#). Industries are classified using the 48 industry classification scheme of [Fama and French \(1997\)](#). In the first two columns, we perform subsample analysis by splitting the full sample into two subgroups according to the politically sensitive industry indicator and then examine the interactive effects between the election year dummy and the politically sensitive industry indicator in the last column. Each column is based on such a subsample, indicated by the column heading. The unit of observation is at firm-year level. The dependent variable in all regressions is R&D intensity, defined as the ratio of R&D expenditure to total assets. The regressions include the same control variables as in the baseline model in Table 2 but the coefficients are omitted from the table to save space. Table A1 in the [appendix](#) defines the variable descriptions as well as the variable sources in detail. The regressions include firm and year fixed effects. Standard errors, reported in parentheses below coefficient estimates, are clustered at the firm level and corrected for heteroscedasticity. To save space, we suppress the estimates of firm-specific and state economics control variables. Data are for the period 1976 to 2020. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 6
Growth opportunities and high-growth firms

	Dependent variable: R&D intensity					
	BFP-high-tech			HJT-high-tech		
	High-tech (1)	Low-tech (2)	Interacted (3)	High-Tech (4)	Low-tech (5)	Interacted (6)
Close election dummy	0.0047*** (0.0017)	−0.0002 (0.0007)	0.0005 (0.0009)	0.0066*** (0.0017)	−0.0004 (0.0009)	0.0008 (0.0009)
Election × BFP-high-tech indicator			0.0041*** (0.0014)			
Election × HJT-high-tech indicator						0.0037** (0.0016)
N	11,643	13,534	25,177	10,063	15,114	25,177
R ²	.8171	.8315	.8447	.835	.8419	.8447
Constant	Yes	Yes	Yes	Yes	Yes	Yes
Firm-specific controls	Yes	Yes	Yes	Yes	Yes	Yes
State economics controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes

This table examines cross-sectional variation of growth opportunities of high-tech versus non-high-tech industries. We consider two classifications of high-tech firms. First, in columns 1–3, we follow [Brown, Fazzari, and Petersen \(2009\)](#) and create an indicator variable, *BFP-High-Tech*, set equal to one for firms operating in the seven high-tech industries, namely, drugs, office equipment and computers, electronic components, communication equipment, scientific and medical instruments, and software, where industries are classified over three-digit SIC code. Unlike other types of investment, R&D investment is highly concentrated in the seven high-tech industries. Second, in columns 4–6, we follow the work of [Hall, Jaffe, and Trajtenberg \(2005\)](#), we consider firms in the pharmaceuticals (13), medical instruments (12), chemicals (14), computers (35), communications (32), and electrical (22) industries that derive most of their value from R&D investments and patents. Industries are classified using the 48 industry classification scheme of [Fama and French \(1997\)](#). We therefore set a *HJT-high-tech* indicator to one if a firm belongs to one of these above-mentioned industries. The dependent variable in all regressions is R&D intensity, defined as the ratio of R&D expenditure to total assets. The regressions include the same control variables as in the baseline model in Table 2, but the coefficients are omitted from the table to save space. Table A1 in the [appendix](#) defines the variable descriptions as well as the variable sources in detail. We use the baseline regression specification and control for firm and year fixed effects. Standard errors, reported in parentheses below coefficient estimates, are clustered at the firm level and corrected for heteroscedasticity. Data are for the period 1976 to 2020. * $p < .1$; ** $p < .05$; *** $p < .01$.

Takeaway. This is the paper’s main mechanism table. When R&D is more clearly a growth option with technical uncertainty and long development lags, the positive effect of uncertainty becomes much stronger.

5.7 Table 7: Dynamics over the four-year election cycle

Read:

- The **close election year** is consistently positive and significant.
- The **preelection** dummy is insignificant.
- The **post-election** dummies are not robustly significant once included jointly.

Takeaway. The R&D response is concentrated in the close election year itself. There is little evidence of pretrends or a strong post-election rebound.

5.8 Table 8: Asset redeployability and substitution tests

Read:

- The interaction between uncertainty and redeployability is **insignificant** in the R&D regression.
- In the capital expenditure regression, higher redeployability dampens the negative election-quarter effect, especially in quarter 3.

Takeaway. Redeployability matters for capital investment in the way prior literature predicts, but it does *not* explain the R&D result. This weakens the simple substitution explanation.

Table 7
Changes in R&D across the gubernatorial election cycle

	Dependent variable: R&D intensity			
	(1)	(2)	(3)	(4)
Preelection dummy (−1)	−0.0003 (0.0009)		0.0014 (0.0022)	
Close election dummy	0.0026** (0.0011)	0.0041*** (0.0011)	0.0054*** (0.0020)	0.0030*** (0.0010)
Post-election dummy (+1)		0.0040** (0.0018)	0.0045 (0.0029)	0.0030 (0.0021)
Post-election dummy (+2)				−0.00007 (0.0017)
N	23,442	23,267	21,645	21,327
R ²	.8533	.8436	.8519	.8426
Constant	Yes	Yes	Yes	Yes
Firm-specific controls	Yes	Yes	Yes	Yes
State economics controls	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

This table investigate the dynamics of R&D investment across the four-year election cycle. In doing this, we follow [Julius and York \(2012\)](#) to estimate the following augmented baseline regression model that also include indicators for pre- and post-election years. The dependent variable in all specifications is R&D intensity, defined as the ratio of R&D expenditure to total assets. Variables of interests are the four election timing dummy variables. Close Election Dummy is the close election year dummy as defined in our baseline regression model. Similar to the election dummy, Preelection dummy is a dummy variable that equals one in the year right before an election year. Post-election dummy (+1) and Post-election dummy (+2) are dummy variables that indicate the first year and second year following a close election year, respectively. In columns 1 and 2, we add the pre- and post-election dummy variables to the baseline regression model separately. In column 3, we include both dummy variables in addition to the election year dummy. In column 4, we only include the two post-election dummies in the baseline regression model. We include firm and year fixed effects in all specifications. To save space, we suppress the estimates of firm-specific and state economics control variables. Table A1 in the appendix defines the variable descriptions as well as the variable sources in detail. Standard errors, reported in parentheses below coefficient estimates, are clustered at the firm level and corrected for heteroscedasticity. Data are for the period 1976 to 2020. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 8
Asset redeployability and sensitivity to political uncertainty

Dependent variable	(1) R&D intensity	(2) Capex/Assets
Close election quarter 1	0.0006 (0.0004)	−0.0006 (0.0014)
Close election quarter 2	0.0012*** (0.0004)	0.0009 (0.0014)
Close election quarter 3	0.0013*** (0.0004)	−0.0057*** (0.0014)
Close election quarter 4	0.0012*** (0.0004)	−0.0020 (0.0015)
CEQ1 × Redeployability	0.0014 (0.0021)	0.0004 (0.0033)
CEQ2 × Redeployability	0.0009 (0.0020)	0.0018 (0.0032)
CEQ3 × Redeployability	0.0024 (0.0021)	0.0126*** (0.0032)
CEQ4 × Redeployability	0.0009 (0.0024)	0.0047 (0.0035)
Redeployability	−0.0304*** (0.0024)	−0.0230*** (0.0023)
N	94,256	94,256
R ²	.2680	.2016
Firm fixed effects	Yes	Yes
Quarter fixed effects	Yes	Yes
Firm controls	Yes	Yes

This table reports R&D and capital expenditure investment regressions using Compustat Quarterly data. The sample is over close election periods (margin of victory is less than 5%). *Close election election-quarter* (CEQ) dummies for quarters 1 through 4 are set equal to one during the specified quarter in the election year if the election is classified as "close." The Redeployability variable is from [Kim and Kang \(2017\)](#). We include firm and year fixed effects in all specifications. Standard errors, reported in parentheses below coefficient estimates, are clustered at the firm level and corrected for heteroscedasticity. Data are for the period 1988 to 2020. * $p < .1$; ** $p < .05$; *** $p < .01$.

5.9 Table 9: State-level EPU index as an alternative proxy

Read:

- The state EPU index rises significantly in **quarters 3 and 4** of close-election cycles, with the largest increase in quarter 4.
- In investment regressions, log(State EPU) has a **positive** coefficient for R&D and a **negative** coefficient for capital expenditures.
- The IV-style estimate using the election-related component of EPU yields a larger R&D coefficient, about **0.0016**.

Takeaway. The main finding is not tied to one uncertainty measure. A newspaper-based uncertainty index leads to the same qualitative conclusion.

Table 9
Alternative proxy for policy uncertainty: State-level EPU index

A. log(State-level EPU index) around close elections					
	Q1	Q2	Q3	Q4	
Close election quarter coefficient	−0.0140 (0.0137)	0.0231 (0.0125)	0.0171*** (0.0055)	0.0653*** (0.0148)	
N	6,566				
R ²	0.8452				
B. Investment regressions					
Dependent variable	(1) R&D intensity	(2) Capex/Assets	(3) R&D intensity	(4) Capex/Assets	(5) R&D intensity
log(State-level EPU)	0.0006*** (0.0001)	−0.0043*** (0.0001)	0.0012*** (0.0003)	−0.0046*** (0.0008)	0.0016** (0.0007)
EPU × Redeployability			−0.0006 (0.0007)	0.0082*** (0.0020)	
Redeployability			−0.0012 (0.0033)	−0.0014** (0.0085)	
Firm controls	Yes	Yes	Yes	Yes	Yes
N	450,714	450,714	362,500	362,500	112,333
R ²	.7182	.4116	.7165	.4754	.1990

This table reports R&D and capital expenditure investment regressions using Compustat Quarterly data with an alternative proxy for policy uncertainty. Panel A reports state-level regressions of the state-level EPU index on close election quarters. The regressions include state and year fixed effects. Standard errors, reported in parentheses, are clustered by state. Panel B reports regression results of R&D intensity and capital expenditures. The variables of interest is the natural logarithm of the state-level EPU index of [Baker, Davis, and Levy \(2022\)](#). We include Table 9's cash flow, firm size, and firm and quarter fixed effects in all specifications. Column 5 reports the estimation results from a two-stage least squares model where close election indicators are used as instruments for the logarithm of the state-level EPU index. Table A1 in the appendix defines the variable descriptions as well as the variable sources in detail. Standard errors, reported in square brackets below coefficient estimates, are clustered at the firm level and corrected for heteroscedasticity. Data are for the period 1988 to 2020. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 10
Political uncertainty and R&D intensity: Robustness tests

A. Robustness tests for the baseline results in Table 2					
	Dependent variable: R&D intensity				
	(1)	(2)	(3)	(4)	(5)
Election dummy	0.0023** (0.0011)	0.0025*** (0.0009)	0.0024** (0.0009)	0.0020*** (0.0009)	0.0026** (0.0007)
N	25,177	18,110	23,870	24,736	15,993
R ²	.8447	.8479	.8500	.8446	.8531
Constant	Yes	Yes	Yes	Yes	Yes
Firm-specific controls	Yes	Yes	Yes	Yes	Yes
State economics controls	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes

(Continued)

Table 10
(Continued)

B. Impact of electoral uncertainty on state government spending		
	Total state expenditures (in millions USD)	Total expenditures to general revenues
	(1)	(2)
Election dummy	−0.7254 (1.6344)	0.0019 (0.0031)
Legislative election	−0.0936 (0.8910)	−0.0073** (0.0030)
Republican governor	−4.4872 (3.1123)	−0.0061 (0.0053)
Split legislature	−2.4263 (2.3874)	−0.0092 (0.0139)
GDP growth	0.3726 (0.2664)	−0.0046*** (0.0008)
Unemployment	1.4841 (1.0219)	0.0131*** (0.0029)
Constant	17.3335*** (5.4464)	1.0665*** (0.0163)
N	2,150	2,150
R ²	.405	.398
State FE	Yes	Yes
Year FE	Yes	Yes

This table presents robustness tests for the baseline R&D intensity regression results shown in Table 2. In panel A of Table 10, we use the baseline R&D regression specification and control for firm and year fixed effects across all columns. Standard errors, reported in parentheses below coefficient estimates, are clustered at the state level and corrected for heteroscedasticity in columns 2 to 5. In column 1, we reestimate the baseline regression model with standard errors triple clustered by state, firm, and year. Column 2 reports regression results based on the subsample of firms based on NYSE, Amex, and Nasdaq only (Chang = 11, 12, 14). In columns 3 and 4, we remove observations in the dot-com bubble period (i.e., 1999–2000) and financial crisis period (i.e., 2007–2008), respectively. Column 5 includes firm headquarters in California, Massachusetts and New York to ensure that our results are not driven by a small number of large states with disproportionate representation of innovative firms in our sample. To save space, we suppress the estimates of firm-specific and state economics control variables. Data are for the period 1976 to 2020. * $p < .1$; ** $p < .05$; *** $p < .01$. In panel B of Table 10, we examine the effects of electoral uncertainty induced by gubernatorial elections on state government spending. The dependent variable is total state expenditures in column 1 and total expenditures as a percentage of general revenues in column 2. Independent variables include additional dummy variables to indicate the timing of gubernatorial elections and state legislative elections, as well as the partisan status of state governors and legislatures. We also account for attributes of state economic conditions, including unemployment rate and gross domestic product (GDP) growth rate. Variable of interest is the election year indicator. In both columns, state and year fixed effects are included in the regression models, and standard errors, reported in parentheses below coefficient estimates, are clustered by state and year.

5.10 Table 10: Robustness and political-business-cycle concerns

Read:

- The baseline election coefficient survives triple clustering, dropping OTC firms, excluding the dot-com years, excluding the financial crisis, and excluding California, Massachusetts, and New York.
- The election dummy is **insignificant** in state-spending regressions.

Takeaway. The result is not driven by a few states, by a few crisis periods, or by election-induced state spending booms.

6 Short summary

- **Method.** Use close U.S. gubernatorial elections as quasi-exogenous shocks to political uncertainty and estimate panel DD regressions of firm-level R&D intensity.
- **Identification.** The design relies on fixed-by-law election timing, staggered state election cycles, firm and time fixed effects, and stronger effects in genuinely close races validated with polling and volatility data.
- **Mechanism.** The evidence favors the **growth-option view**: R&D is more likely to rise under uncertainty when projects are politically sensitive, high-tech, technically uncertain, or multi-stage.
- **Main result.** Political uncertainty increases R&D but decreases capital expenditures. The effect is economically meaningful and strongest in close elections and in industries where R&D is most option-like.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper shows that political uncertainty does **not** have a universally negative effect on firm investment.
- Using close gubernatorial elections, it finds that uncertainty **raises R&D** even though it lowers capital expenditures.
- The evidence points to a model in which R&D behaves like a **growth option** rather than a standard irreversible investment.

7.2 Economic intuition

- Physical capital is often hard to reverse, so uncertainty makes waiting attractive.
- R&D, by contrast, can be exploratory. Firms invest now to learn, then either continue if conditions improve or abandon if conditions are bad.

- Because the downside is truncated but the upside expands with uncertainty, the incentive to start R&D can rise even while the incentive to commit to physical capital falls.

7.3 Takeaway

The main lesson is that the real effects of uncertainty are **investment specific**. For policymakers, this means that broad claims such as “political uncertainty hurts investment” are incomplete. Political uncertainty can indeed reduce traditional capital formation, but it can simultaneously stimulate exploratory and innovation-oriented activity such as R&D. That is the paper’s central “bright side” result.